

Davian R. Chin

STEM Educator — Mathematics, Computer Science & Physics

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Teaching Profile

Educator and mathematical scientist with 15+ years' teaching and leadership experience across the UK, China, Kazakhstan and Jamaica (secondary to A level/CAPE). Experienced in curriculum design (CSEC/CAPE/A level), department leadership, and coaching teachers in evidence-informed practice.

- MA Education research: developed and evaluated a computational-thinking framework for mathematics teaching.
- Converts advanced STEM ideas into classroom-ready resources: Python/Jupyter lesson libraries, lab-based modules with rubrics, and interactive demonstrators.
- Current PhD research (fractional calculus, stochastic processes, scientific machine learning) strengthens subject knowledge across mathematics, physics and computer science.

Seeking a teaching position in mathematics, computer science, or physics.

Teaching & Leadership Experience

Founder & Director, Neuryte Learning Ltd

STEM education & AI consultancy – 2024–present

- Consult on the design and delivery of STEM education programmes (mathematics, science, computer science, engineering, AI/ML engineering) for schools, colleges, education providers and edtech organisations.
- Lead end-to-end STEM project development and management (scoping, design, delivery, evaluation), including practical AI integration.
- Teach and tutor online and face-to-face across mathematics, science, computer science and physics; develop reproducible computational resources in Python (notebooks, demonstrators).

Teacher of Computer Science

Walton High, Milton Keynes, UK – Oct 2023–Dec 2024

- Taught computer science to secondary students preparing for external examinations; built skills in computational thinking, modelling, design, optimisation and data-informed decision-making.
- Delivered learning through guided inquiry, team-based real-world projects and hands-on skill practice.

Teacher of Mathematics & Science

The Bedford Academy, Bedford, UK – Sep 2019–Aug 2023

- Taught mathematics and science through real-world, application-led examples; used technology including simulation and interactive visualisation.
- Raised progress for lower-attaining learners through a hands-on, context-based approach; increased STEM engagement through national science competitions.

Mathematics Department Chair & Lead Teacher

Shenzhen Vanke Meisha Academy, China – Aug 2016–Jul 2019

- Led a multicultural team to build the mathematics department from scratch — strategic focus, communication protocols and operational framework.
- Served on the school's strategic leadership team (curriculum and assessment, school improvement, residential school committees).
- Carried out action research on integrating computational thinking and cross-curricular frameworks into mathematics and science teaching.

STEM/CLIL Mathematics Teacher & Instructional Designer

Nazarbayev Intellectual Schools, Kazakhstan

– Aug 2014–Jul 2016

- Taught Cambridge Integrated Mathematics to Grades 11–12 for university entry; supported English-medium learning for Russian- and Kazakh-speaking students and staff.
- Delivered masterclasses on effective teaching, learning and assessment; modelled interdisciplinary, problem-based approaches.
- Led implementation and evaluation of a summer STEM school on mathematical applications in biodiversity and rocket science.

A-Level Mathematics & Physics Teacher & Programmes Coordinator St Hugh's High School, Ministry of Education, Jamaica – Sep 2009–Aug 2014

- Designed curriculum and instructional plans for CSEC, CAPE and A-Level mathematics and physics courses; taught grades 11–13 in preparation for university entry.
- Supervised small mathematics research projects and physics laboratory work; assisted with content and teaching-strategy training for new teachers.

Mathematics Teacher (Grades 10–11)

Gaynstead High School, Jamaica – Jan–Aug 2009

Curriculum & lesson design

Open-source teaching materials – ongoing

- **comp_math** — A-Level mathematics lesson library taught through Python and Jupyter notebooks (rational and nonlinear functions and their graphs, the binomial theorem with tested implementations, statistics and data summarisation).
- **differentiable-probabilistic-programming-curriculum** — a full curriculum spine for differentiable and probabilistic programming (JAX/NumPyro): modules with prerequisites and learning outcomes, labs, notebooks, capstone projects, assessment rubrics and machine-validated curriculum metadata.
- **Doctoral-era teaching materials** — guided learning manuals and workbooks on mathematical methods, scientific machine learning and neural networks, signal processing and wavelets, non-Gaussian stable laws, and fractional reservoir computing; plus a scheme of work for a fractional-calculus/long-range-dependence study week.

Educational writing & public engagement

2014–present

- Educational writing and reflection on modern pedagogy, including the essay *A thought on teaching students in the modern world* (2014).
- Public engagement with research: conference talks and posters communicating mathematics and critical brain science to non-specialist academic audiences.

Subjects & Subject Knowledge

Mathematics: real and complex analysis, calculus, linear algebra, ordinary and partial differential equations, probability and statistics, number theory, discrete and decision mathematics, mathematical modelling.

Physics: classical dynamics, heat/diffusion and wave PDEs, statistical physics, dynamical systems and criticality, computational and numerical modelling.

Computer Science: programming (Python primary; also C/C++, R, MATLAB, SQL, Haskell), software engineering, databases and networks, artificial intelligence and machine learning, data science, cloud and big-data fundamentals.

Educational Research & Writing

- MA Education dissertation (University of South Wales, 2018): *Strategies towards a more visible, practical and culturally relevant mathematics education framework: teaching mathematics through computational thinking* — comparative analysis of formative and summative assessment evidence showing improved outcomes under the computational-thinking approach.
- Author of research-adjacent teaching materials: interactive neural-network visualiser for teaching deep learning (neural-net-model, JavaScript), and the lesson libraries and workbooks above.

Education

PhD, Biomedical Engineering (Computational Neuroscience)
(expected 2028)

University of Reading, UK – 2025–present

Research-grade specialisation underpinning subject knowledge: fractional calculus, stochastic processes, functional analysis, machine learning.

MSc, Computer Science (AI, ML & Big Data Technologies)

University of East London, UK – 2023

MA, Education (Development, Management & Technology)

University of South Wales, UK – 2018

BSc, Mathematics & Computer Science

University of the West Indies, Mona, Jamaica – 2008

Graduate coursework and research in mathematics and theoretical physics (UWI Mona, 2010–2013).

ICT & Technical Skills

Teaching technology: Python, Jupyter notebooks, interactive demonstrators, LaTeX for teaching materials, Google Colab and cloud classrooms, Git/GitHub for versioned coursework, automated curriculum validation (JSON Schema, CI).

Subject software: NumPy/SciPy, pandas, scikit-learn, JAX, PyTorch, Matplotlib, MATLAB, R, SQL.

Systems: Linux, Docker basics, Google Cloud, high-performance computing.

Certifications & Professional Development

- PyTorch Essential Training: Deep Learning (LinkedIn Learning, 2022).
- Python Programmer Track (DataCamp, 2020); Python for Data Science: Fundamentals & Intermediate (Dataquest, 2019).
- Computer Science Career Path & Code Foundations (Codecademy, 2020).
- Fellow, Institute of Mathematics and its Applications (FIMA).

References available on request.